

```
sudo apt update
sudo apt install apt-transport-https ca-certificates curl software-properties-common -y
sudo apt install docker.io -y
sudo systemctl start docker
sudo systemctl enable docker
docker --version
sudo docker run hello-world
mkdir my-python-app
cd my-python-app
gedit app.py
```

APP.PY:

```
from flask import Flask
app = Flask(__name__)
@app.route('/')
def home():
    return "Hello Docker is running this python app succesfully"
if __name__ == "__main__":
    app.run(host='0.0.0.0', port=5000)
```

gedit Dockerfile

DOCKERFILE:

```
FROM python:3.9-slim
WORKDIR /app
COPY . .
RUN pip install -r requirements.txt
EXPOSE 5000
CMD ["python", "app.py"]
```

gedit requirements.txt

REQUIREMENTS:

flask

```
sudo systemctl start docker
```

```
docker build -t my-app-image .
```

```
docker images
```

```
docker run -d -p 5000:5000 --name myapp-container my-app-image
```

```
docker ps
```

```
git init
```

```
git add .
```

```
git commit -m "Initial Commit"
```

```
git branch -M main
```

```
git remote add origin https://github.com/KhushiJigali/docker.git
```

```
git push -u origin main
```

JENKINS FILE:

```
pipeline {
```

```
    agent any
```

```
    environment {
```

```
        // Jenkins Credentials ID
```

```
        DOCKERHUB_CREDENTIALS = 'dockerhub-creds'
```

```
        // Your Docker Hub username/repo
```

```
        APP_NAME = "khushijigali/dockerexperiment"
```

```
    }
```

```
    stages {
```

```
        stage('Checkout') {
```

```
            steps {
```

```
                checkout scm
```

```

    }
}

stage('Build Docker Image') {
    steps {
        script {
            sh "docker build -t ${APP_NAME}:${BUILD_NUMBER} ."
        }
    }
}

stage('Login and Push') {
    steps {
        withCredentials([
            usernamePassword(
                credentialsId: "${DOCKERHUB_CREDENTIALS}",
                passwordVariable: 'PASS',
                usernameVariable: 'USER'
            )
        ]) {

            sh "echo ${PASS} | docker login -u ${USER} --password-stdin"

            sh "docker push ${APP_NAME}:${BUILD_NUMBER}"

            sh "docker tag ${APP_NAME}:${BUILD_NUMBER} ${APP_NAME}:latest"

            sh "docker push ${APP_NAME}:latest"
        }
    }
}
}

```